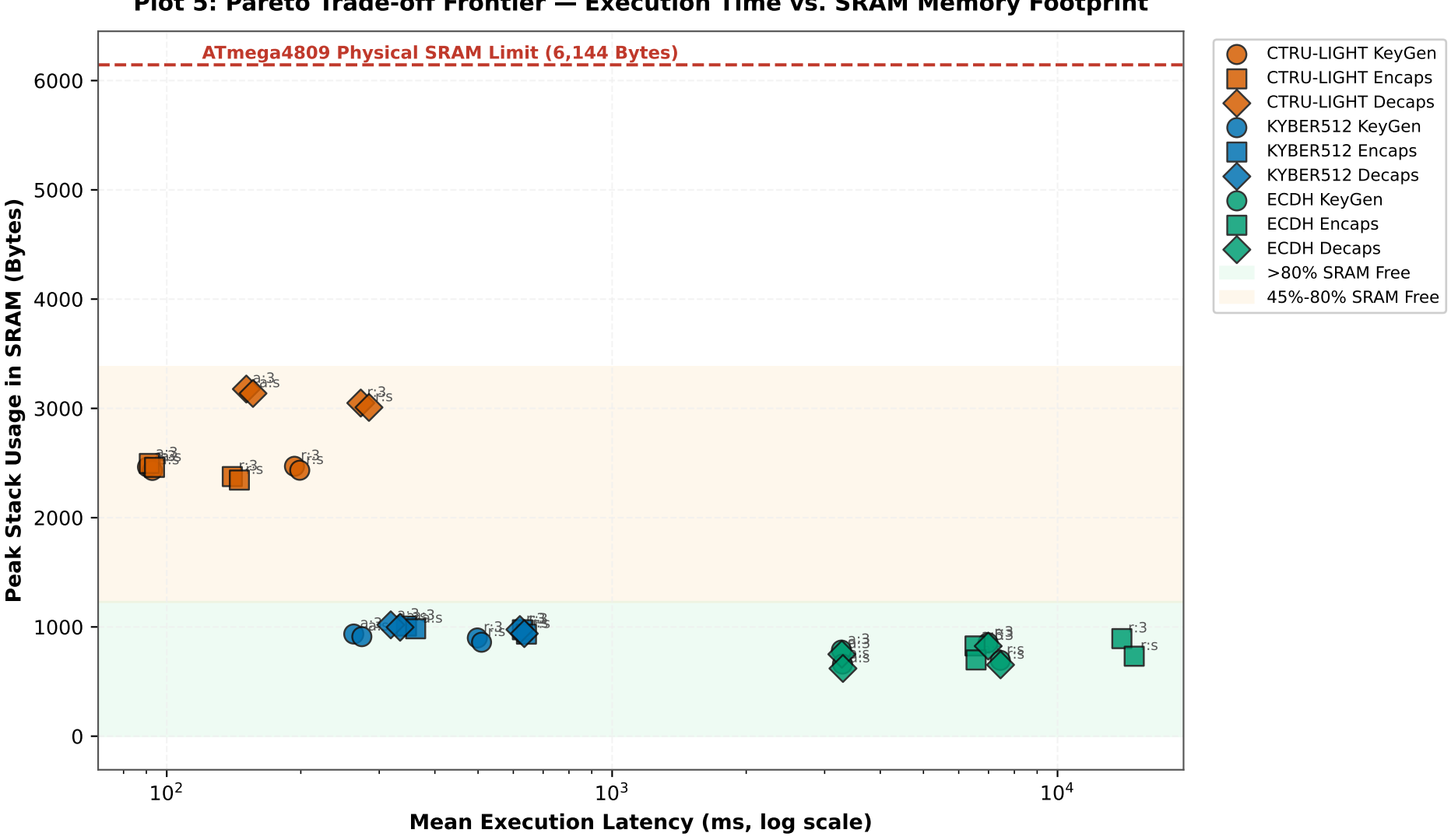
Plot 5: Pareto Trade-off Frontier — Execution Time vs. SRAM Memory Footprint
ATmega4809 Physical SRAM Limit (6,144 Bytes)
Peak Stack Usage in SRAM (Bytes)
Mean Execution Latency (ms, log scale)
0
1000
2000
3000
4000
5000
6000
10^2
10^3
10^4
CTRU-LIGHT KeyGen
CTRU-LIGHT Encaps
CTRU-LIGHT Decaps
KYBER512 KeyGen
KYBER512 Encaps
KYBER512 Decaps
ECDH KeyGen
ECDH Encaps
ECDH Decaps
>80% SRAM Free
45%-80% SRAM Free